Hydropower Site Suitability

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Contents

1 Introduction

This document describes the methodology, data processing pipeline, model training, and evaluation for the project focused on predicting site suitability for new hydropower plants using satellite imagery and deep learning segmentation models.

The objective is to identify optimal locations for new hydropower plants by analyzing a variety of environmental factors using supervised learning. While there is limited available research that directly uses machine learning models in the analysis of site suitability for hydropower sites, our method, with the use of satellite imagery, has the advantage that it can potentially uncover suitable locations that may not be considered by typical manual or rule-based analysis.

We decided to use two approaches, resulting in two ML pipelines. The two pipelines start the same by making requests to the Sentinel Hub API to acquire the satellite data. The first approach models the gathered data in a tabular format and uses traditional ML algorithms (Random Forest, XGBoost) to make predictions. The second approach models the data as images with multiple channels and uses a CNN based on resnet to classify the images.

2 Data Sources and Gathering

The resulting dataset consists of about 3000 positive samples consisting of locations of hydropower plants from Europe and 5000 negative samples of random locations from European rivers.

2.1 Satellite Data

Using the Sentinel Hub API, we gather Sentinel-2 satellite imagery for the regions of interest. We focus on bands B3, B4, B8 and B11, which cover the visible and near-infrared spectrum.

The following spectral indices are calculated:

- NDVI (Normalized Difference Vegetation Index) - used to assess vegetation health and density. Vegetation cover can be used to assess land suitability for hydropower plants.
- NDWI (Normalized Difference Water Index) - used to identify water bodies and assess their extent. Water detection is important in identifying potential sites for hydropower plants.
- NDBI (Normalized Difference Built-up Index) - used to detect built-up areas and urbanization. This helps in avoiding locations that are too close to urban areas.
- MNDWI (Modified Normalized Difference Water Index) - used to enhance and detect water features.

2.2 Topographic Data

Topographic data is obtained from the Copernicus Digital Elevation Model (DEM). The DEM provides elevation data which is crucial for assessing the potential energy generation capacity of a site and the potential of forming and accumulation lakes near the plants. From the DEM, we also derive slope data, which helps in understanding the terrain and its suitability for hydropower infrastructure.

2.3 Hydrological and Precipitation Data

Hydrological data, specifically river discharge data, is obtained from the Hydrorives database and precipitation data is gathered using the Open-Meteo API. River discharge is a key factor in determining the potential energy output of a hydropower plant, while precipitation data helps in understanding the water availability in the region.

2.4 Labels

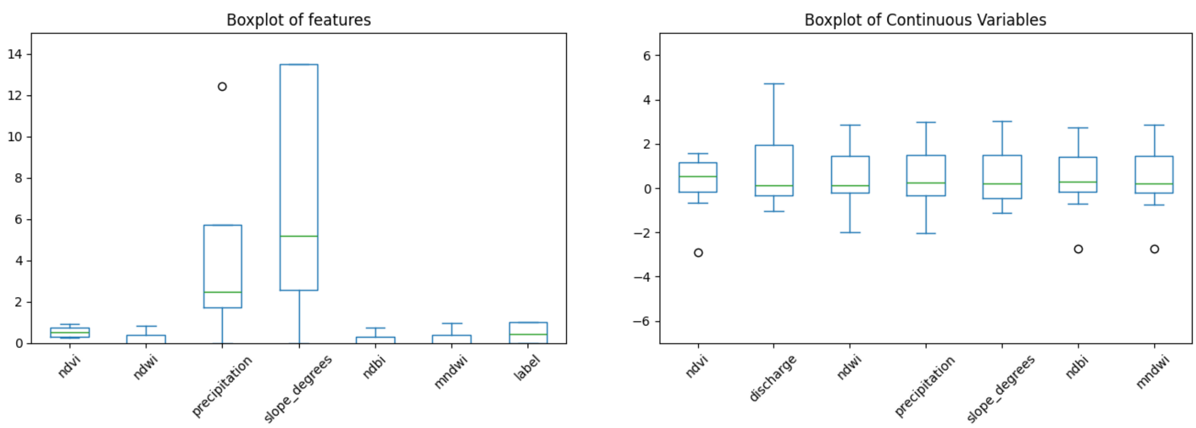
Positive samples: near existing hydropower plants.
Negative samples: random locations at least 30 kilometers away from any hydropower plant.

3 Data Preprocessing

3.1 Preprocessing for Tabular Data

Preprocessing steps:

- Splitting data: Performed train-test split (e.g., 80-20).
- Handling missing values: Imputed missing values using mean strategy.
- Outlier detection: Identified and treated outliers using IQR method and replaced them using the same imputation strategy as for the missing values.
- Normalization: Scaled features to a standard range.



(a) Boxplot before preprocessing

(b) Boxplot after preprocessing

Figure 1: Comparison of feature distributions before and after preprocessing.

3.2 Preprocessing for Image Data

Feature cube construction: shape (H, W, C) with channels for spectral indices, DEM, slope, discharge, precipitation. The 8-channel feature cubes consist of the following spectral indices and geophysical parameters:

1. NDVI (Normalized Difference Vegetation Index)
2. NDWI (Normalized Difference Water Index)
3. NDBI (Normalized Difference Built-up Index)
4. MNDWI (Modified Normalized Difference Water Index)
5. DEM (Digital Elevation Model)
6. Slope
7. Discharge
8. Precipitation

When creating the Dataset for training the custom weights were assigned to the channels of the image to better reflect the importance of the feature represented by the channel.

Table 1: Channel Weights Used for Image Feature Cubes

Channel	NDVI	NDWI	NDBI	MNDWI	DEM	Slope	Discharge	Precipitation
Weight	0.8	1.8	0.6	2.0	1.5	2.2	2.4	1.0

The data set was then slipt into 3 datasets:

- Train - 70% of the data.
- Validation - 20% of the data.
- Test - 10% of the data.

4 Machine Learning Models

4.1 Tabular Data Models

For the tabular representation of the dataset, we trained and compared three widely used ensemble learning algorithms:

- **Random Forest**: An ensemble of decision trees using bootstrap sampling and feature bagging. It provides robust performance, reduces overfitting, and allows for feature importance ranking.
- **MLP (Multi-Layer Perceptron)**: A feedforward neural network with multiple layers and non-linear activation functions. It can model complex relationships in the data and is suitable for tabular data.

- **XGBoost**: An optimized implementation of gradient boosting decision trees. It generally provides higher accuracy, faster convergence, and better handling of imbalanced data.

4.2 CNN Classification Model

The CNN model is based on the pre-trained ResNet-34 architecture, modified to handle the 8-channel input feature cubes. The key modifications brought to the model include:

- **Input Layer Modification**: The first convolutional layer (`conv1`) was modified from the standard 3-channel RGB input to accept 8-channel input:

$$\text{conv1} : \mathbb{R}^{8\times H\times W} \to \mathbb{R}^{64\times H'\times W'} \tag{1}$$

  with kernel size 7x7, stride 2, and padding 3.

- **Feature Extraction**: The backbone ResNet-34 layers (conv2_x through conv5_x) remain unchanged, providing hierarchical feature extraction with skip connections.

- **Classification Head**: The final fully connected layer was modified for binary classification:

$$\text{fc} : \mathbb{R}^{512} \to \mathbb{R}^{2} \tag{2}$$

5 Evaluation and Results

5.1 Tabular Data Models

5.1.1 Random Forest

Hyperparamters used for Random Forest:

- `n_estimators`: 100
- `max_depth`: 30
- `min_samples_leaf`: 2
- `criterion`: 'entropy'
- `class_weight`: 'balanced'
- `max_samples`: 0.8
- `max_features`: 0.5
- `random_state`: 42
- `n_jobs`: -1

Table 2: Random Forest Classification Report

Class	Precision	Recall	F1-score	Support
0 (Unsuitable)	0.90	0.93	0.92	981
1 (Suitable)	0.88	0.83	0.86	614
Accuracy		0.89		1595
Macro avg	0.89	0.88	0.89	1595
Weighted avg	0.89	0.89	0.89	1595

5.1.2 MLP

Hyperparamters used for MLP:

- `hidden_layer_sizes`: (128, 64)
- `activation`: 'relu'
- `solver`: 'adam'
- `learning_rate_init`: 0.001
- `max_iter`: 50
- `batch_size`: 128
- `early_stopping`: True
- `alpha`: 0.001
- `random_state`: 42
- `verbose`: True

Table 3: MLP Classification Report

Class	Precision	Recall	F1-score	Support
0 (Unsuitable)	0.89	0.91	0.90	981
1 (Suitable)	0.85	0.82	0.83	614
Accuracy		0.87		1595
Macro avg	0.87	0.87	0.87	1595
Weighted avg	0.87	0.87	0.87	1595

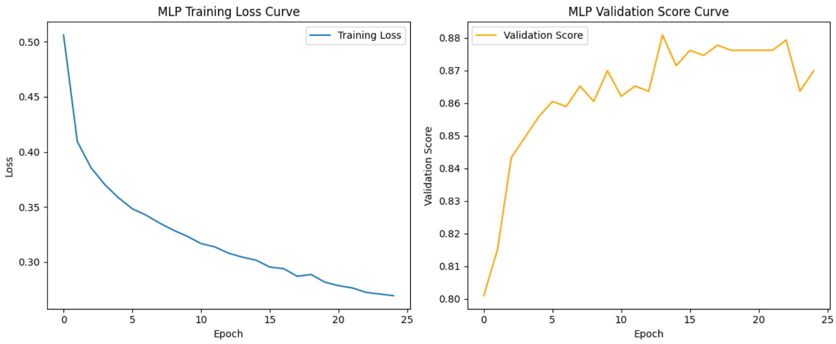


Figure 2: MLP validation accuracy and loss curves over training epochs.

5.1.3 XGBoost

Hyperparamters used for XGBoost:

- `n_estimators`: 300
- `max_depth`: 6
- `learning_rate`: 0.1
- `subsample`: 0.8
- `colsample_bytree`: 0.8
- `random_state`: 42
- `eval_metric`: 'logless'

Table 4: XGBoost Classification Report

Class	Precision	Recall	F1-score	Support
0 (Unsuitable)	0.83	0.94	0.88	981
1 (Suitable)	0.88	0.70	0.78	614
Accuracy		0.85		1595
Macro avg	0.86	0.82	0.83	1595
Weighted avg	0.85	0.85	0.84	1595

5.2 CNN Classification Model

The model was trained over **20 epochs** with the following results:

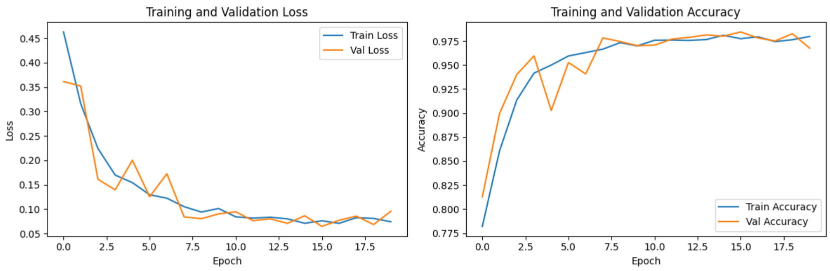


Figure 3: CNN training and validation accuracy/loss curves over epochs.

Table 5: CNN Model Test Results

Metric	Value
Loss	0.0722
Accuracy	0.9776
Precision	0.9784
Recall	0.9776
F1-Score	0.9774

Table 6: CNN Model Confusion Matrix

	Predicted 0	Predicted 1
Actual 0	500	0
Actual 1	18	285

For further validation an interactive map of a zone with already built hydropower plants and many rivers was created to test the model. The map consists of a bounding box that is split into smaller pictures, each one classifed by the model as follows:

Table 7: Interactive Map Classification Summary

Parameter	Value
Grid Shape	72×177
Total Cells	12,744
Classified Cells	12,744
Class Distribution	Hydro Potential: 878
	No Hydro: 11,866
Average Confidence	0.930

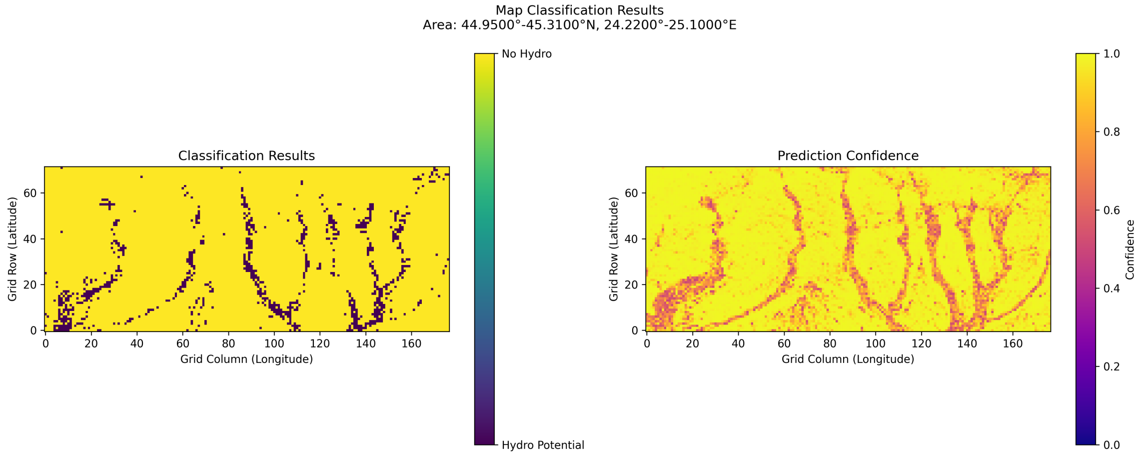


Figure 4: Simple map showing the classification results for hydropower site suitability.

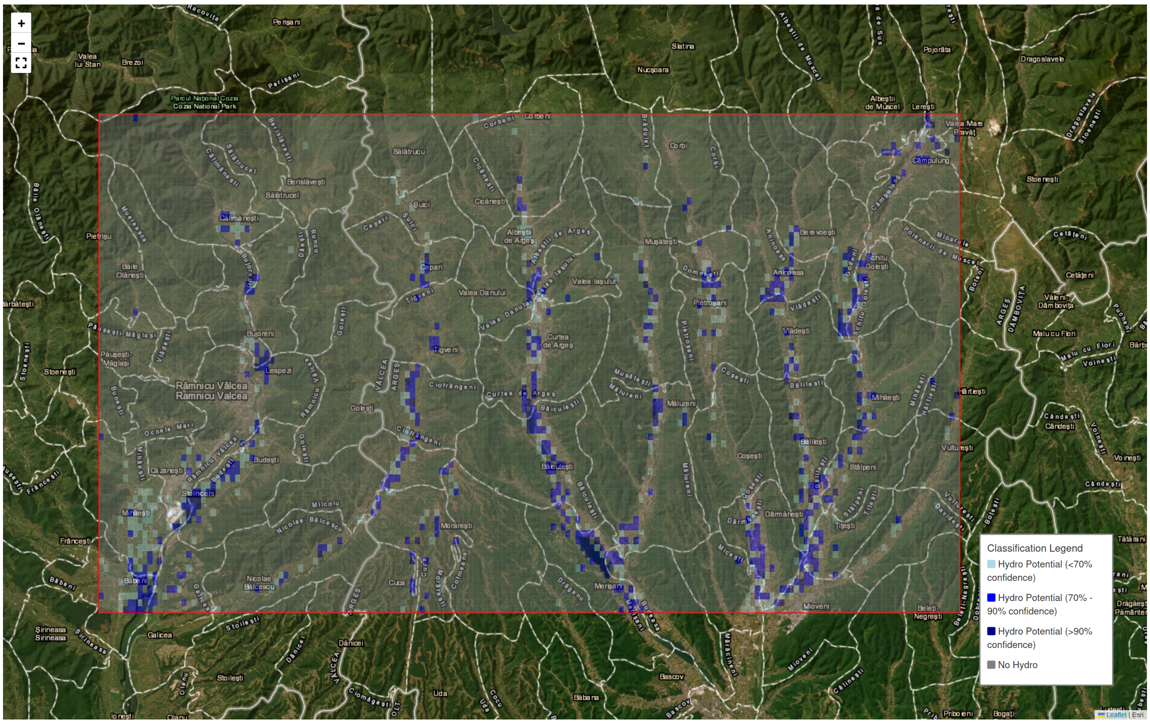


Figure 5: Screenshot of the interactive map used for model validation. The map displays classified regions with hydro potential and unsuitable locations, allowing for visual inspection and further analysis. Each cell represents a classified region, with color coding for hydro potential and unsuitable locations.

5.2.1 Feature Analysis

To better understand the decision-making process of our CNN model, we analyzed several representative examples of locations classified as suitable for hydropower development.

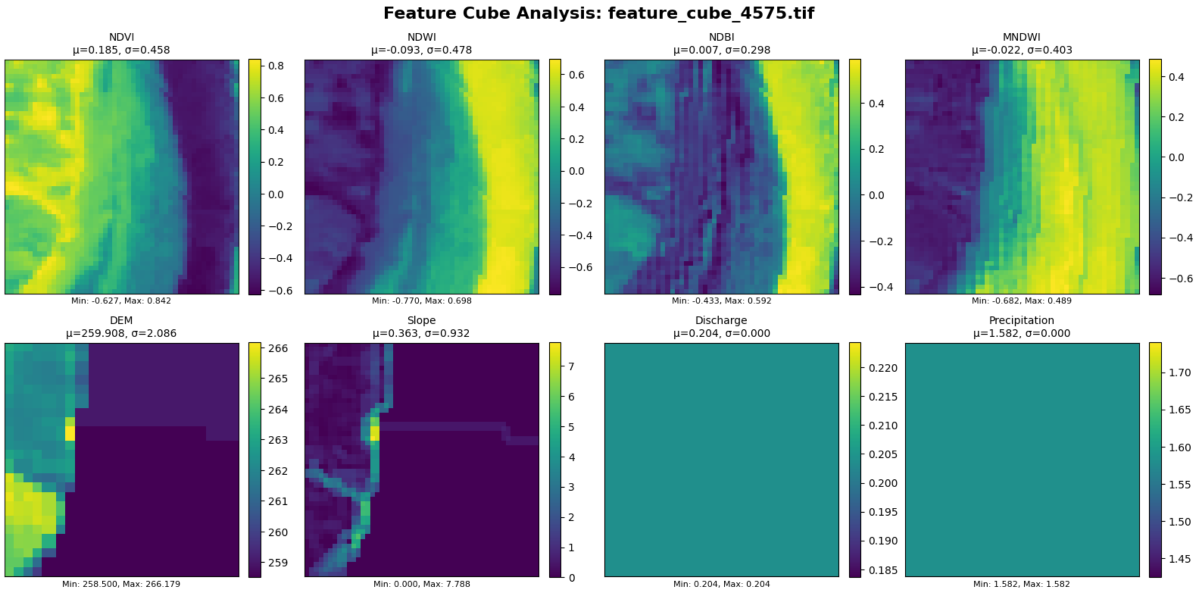


Figure 6: Example 1: Location classified as suitable for hydropower development.

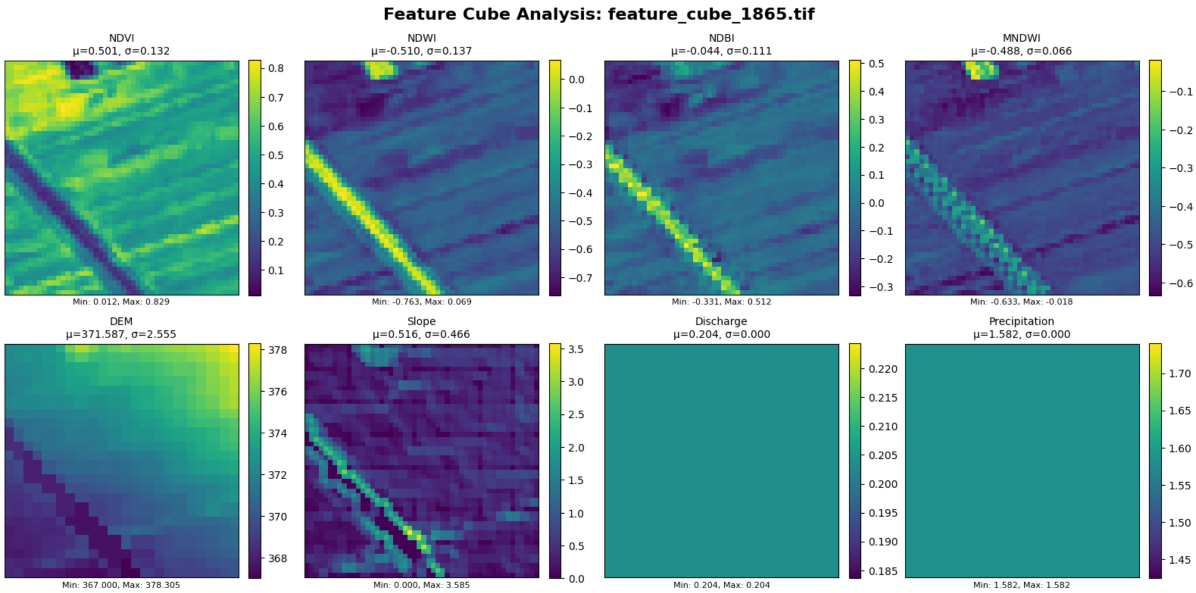


Figure 7: Example 2: Location classified as suitable for hydropower development.

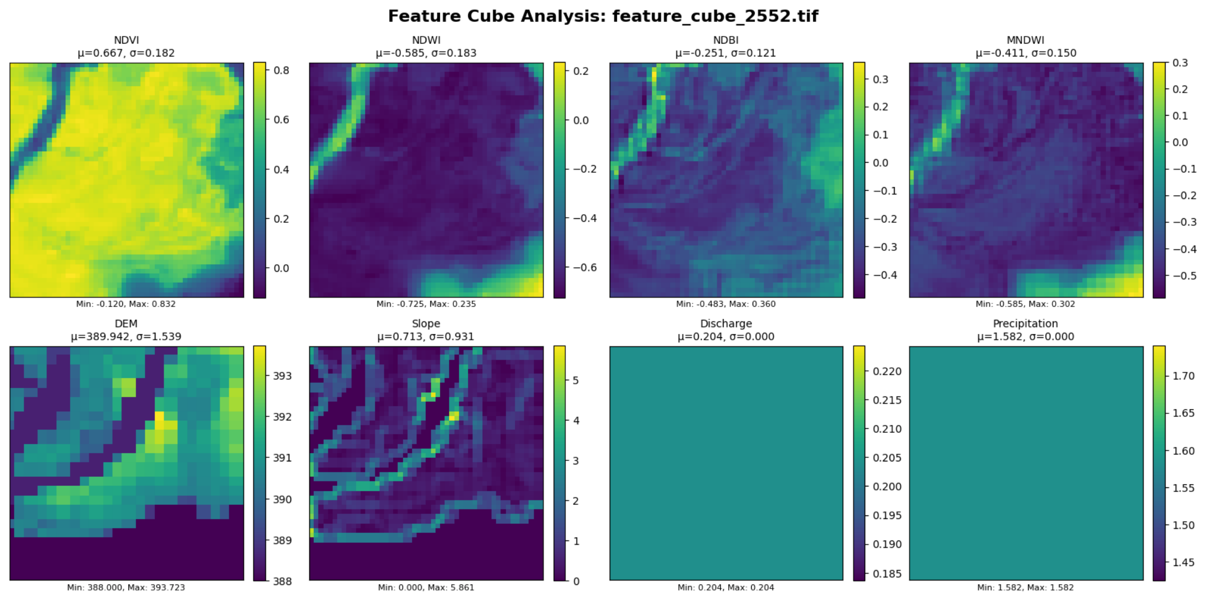


Figure 8: Example 3: Location classified as suitable for hydropower development.

Common Patterns in Positive Classifications

Figures 6, 7, and 8 are some examples that the CNN model classified as suitable hydropower sites.

- **Distinctive Linear Water Features:** All three positive examples exhibit pronounced linear patterns in their water-related spectral indices, particularly visible in the NDWI and MNDWI channels. These linear features represent well-defined river corridors or water channels that are essential for hydropower development.
- **Optimal Topographic Characteristics:** The positive examples consistently show moderate elevation ranges with sufficient topographic variation to provide natural head for power generation. Despite different absolute elevations, all sites maintain moderate slope characteristics that indicate gentle to moderate gradients suitable for run-of-river installations without requiring extensive earthworks.
- **Minimal Built-up Development:** All positive examples demonstrate consistently low NDBI values, confirming minimal existing development.

6 Conclusion

6.1 CNN vs Traditional Models

The results demonstrate that the CNN based approach significantly outperformed the traditional tabular models, achieving a test accuracy of 97.76% compared to the Random Forest's 89% accuracy. This substantial performance difference highlights the advantage of treating the problem as an image classification task, where spatial relationships and contextual information can be better captured through convolutional layers.

6.2 Model Limitations

The CNN model has a tendency toward false positives, particularly in open fields areas, as seen from the testing presented with the interactive map. Adjusting the weights of the input channels seemed to reduce this behaviour, but it is still not prefect and flase positives are still present. This limitation could be explained by the relatively small training dataset size of approximately 8,000 samples (3,000 positive and 5,000 negative).

Additionally, the feature extraction and selection process was conducted with limited guidance from existing research papers on machine learing use in hydropower site suitability. The choice of spectral indices (NDVI, NDWI, NDBI, MNDWI), topographic parameters (elevation, slope), and environmental variables (discharge, precipitation) was based on general principles of environmental analysis. This somewhat arbitrary feature selection may have resulted in suboptimal representation of the atributes needed for hydropower site assessment.

6.3 Future Work and Imporvements

Several directions for future research and development emerge from this work:

- Dataset Expansion: Collecting additional training samples, particularly from diverse geographical regions and environmental conditions, would likely improve model robustness and reduce false positive rates.
- Feature Engineering: Incorporating additional environmental parameters such as seasonal flow variations, soil stability indices, and proximity to existing infrastructure could enhance prediction accuracy.
- Temporal Dynamics: Integrating time-series satellite data to capture ong-term environmental variations would provide a more comprehensive assessment.

6.4 Final Statements

The integration of machine learning techniques with traditional site assessment methodologies offers the potential to significantly improve the efficiency and effectiveness of renewable energy development planning, contributing to more sustainable energy infrastructure deployment.